Design - I

$V_{ov,n} = V_{GS,n} - V_{TH0,n} = 0.48V - 0.33V = 0.15V$

$V_{ov,p} = |V_{GS,p}| - |V_{TH0,p}| = 0.60V - 0.35V = 0.25V$

$\lambda_n = 0.12 + \frac{0.065}{L_n} = 0.12 + \frac{0.065}{0.4} = 0.2825 V^{-1}$

$\lambda_p = 0.06 + \frac{0.04}{L_p} = 0.06 + \frac{0.04}{0.4} = 0.1600 V^{-1}$

$I_{D,n} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W_n}{L_n}\right) V_{ov,n}^2 (1 + \lambda_n V_{OUT,DC})$

$= \frac{1}{2} 480 \times 10^{-6} \times \frac{18.5}{0.4} \times (0.15)^2 (1 + 0.2825 \times \frac{1}{2})$

$= 285.03 \mu A$

$I_{D,p} = \frac{1}{2} \mu_p C_{ox} \left(\frac{W_p}{L_p}\right) V_{ov,p}^2 (1 + \lambda_p (V_{DD} - V_{OUT,DC}))$

$= \frac{1}{2} (150 \times 10^{-6}) \left(\frac{21.3}{0.4}\right) (0.25)^2 (1 + 0.1600 \times \frac{1}{2})$

$= 269.58 \mu A$

$I_D \approx \frac{269.58 + 285.03}{2} = 277.30 \mu A$

$g_{mn} = \frac{2 I_{D,n}}{V_{ov,n}} = \frac{2 \times 285.03 \times 10^{-6}}{0.15} = 3.8 mS$

$g_{mp} = \frac{2 I_{D,p}}{V_{ov,p}} = \frac{2 \times 269.58 \times 10^{-6}}{0.25} = 2.157 mS$

$$r_{on} = \frac{2I_{DP}}{\lambda_n I_{D_n}} = \frac{1}{0.2825 \times 285.05 \times 10^{-6}} = 12.92\,k\Omega$$

$$r_{op} = \frac{1}{\lambda_p I_{DP}} = \frac{1}{0.1600 \times 269.58 \times 10^{-6}} = 23.18\,k\Omega$$

$$r_{out} = r_{on} \| r_{op} = \frac{12.92 \times 23.18}{12.92 + 23.18} = 8.087\,k\Omega$$

$$A_{V0} = -g_m r_{out} = -(3.800 \times 10^{-3})\, 8.087 = -30.73\,V/V$$

$$(29.75\,dB)$$

$$\alpha = \frac{R_0}{R_1 + R_0} = \frac{68.99k}{2k + 68.99k} = 0.9718$$

$$A_V = \alpha \times A_{V0} = 0.9718 \times (-30.737) = -29.87\,V/V$$

$$|A_V|_{dB} = 29.50\,dB \geq 26.02\,dB$$

$$f_L = \frac{1}{2\pi (R_1 + R_0) C_0} = \frac{1}{2\pi (2k + 68.99k) \times 2.5pF}$$

$$= 896.8\,kHz \approx 0.90\,MHz$$

$$f_L \leq 1\,MHz$$

$C_{ov,n} = 0.35 \times 18.5 = 6.48$

$C_{ox,n} = 18.5 \times 0.4 \times 18 = 133.20$ fF

$C_{gs,n} = \frac{2}{3}(133.20) + 6.48 = 95.28$ fF

$C_{gd,n} = C_{ov,n} = 6.48$ fF

$C_{gd,p} = 0.55 \times 21.3 = 7.46$ fF

$C_{db,p} = 0.5 \times 21.3 = 10.65$ fF

$C_{out,tot} = C_{gd,n} + C_{db,n} + C_{gd,n} + C_{db,p} + C_L + C_o$

$= 234.90$ fF

$f_{p,out} = \frac{1}{2\pi \cdot r_{out} \cdot C_{out,tot}} = \frac{1}{2\pi \times 8.087k \times 234.9f}$

$= 83.78$ MHz

$C_{in,miller} = C_{gs,n} + C_{gd,n}(1 + |A_v|)$

$= 298.15$ fF

$f_{p,in} = \frac{1}{2\pi R_{in,eq} C_{in,miller}} = \frac{298.15 \text{ fF}}{}$ 27.37 MHz

$f_H = \frac{1}{\sqrt{\frac{1}{f_{p,in}^2} + \frac{1}{f_{p,out}^2}}}$ = ~~102.21 MHz~~

$= 102.21$ MHz

Design - 2

$\Rightarrow V_{BIASN} = V_{DD} \times \frac{RN_2}{RN_1 + RN_2} = 1 \times \frac{45k}{45k + 55k} = 0.45V$

$\Rightarrow V_{BIASP}$ (V_b in simulation) $= \frac{V_{DD} \times RP_2}{RP_1 + RP_2} = 1 \times \frac{90k}{60k + 90k} = 0.90V$

$\Rightarrow V_{OV,n} = V_{GSn} - V_{TH,n} = 0.45 - 0.33V = 0.12V$

$\Rightarrow V_{OV,P} = |V_{GS,P}| - |V_{TH,P}| = 0.60V - 0.38V = 0.25V$

$\Rightarrow \lambda_n = \frac{0.12 + 0.065}{L_n} = \frac{0.12 + 0.065}{0.3} = 0.3367V^{-1}$

$\Rightarrow \lambda_p = 0.06 + \frac{0.04}{L_p} = 0.06 + \frac{0.04}{0.4} = 0.16V^{-1}$

$\Rightarrow I_{D,n} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W_n}{L_n}\right) V_{OV,n}^2 (1 + \lambda_n V_{OUT,DC})$

$= \frac{1}{2} (400 \times 10^{-6}) \left(\frac{20.6}{0.3}\right) (0.12)^2 \left(1 + 0.3367 \times \frac{1}{2}\right)$

$\approx 280.0 \mu A$

$\Rightarrow I_{D,P} = \frac{1}{2} (150 \times 10^{-6}) \left(\frac{22.5}{0.4}\right) (0.25)^2 \left(1 + 0.16 \times \frac{1}{2}\right)$

$= 284.4 \mu A$

$$\rightarrow g_{mn} = \frac{2I_{D,n}}{V_{ov,n}} = \frac{2 \times 280 \times 10^{-6}}{0.12} = 4.667\ mS$$

$$\rightarrow g_{mp} = \frac{2I_{D,p}}{V_{ov,p}} = \frac{2 \times 289.8 \times 10^{-6}}{0.25} = 12.278\ mS$$

$$\rightarrow r_{on} = \frac{1}{\lambda_n I_{D,n}} = \frac{1}{0.3367 \times 280 \times 10^{-6}} = 10.61\ k\Omega$$

$$\rightarrow r_{op} = \frac{1}{\lambda_n I_{Dp}} = \frac{1}{0.16 \times 289.8 \times 10^{-6}} = 21.95\ k\Omega$$

$$\rightarrow r_{out} = r_{on} \| r_{op} = \frac{10.61 \times 21.95}{10.61 + 21.95} = 7.152\ k\Omega$$

$$\rightarrow A_{vo} = -g_{mn} r_{out} = -(4.667 \times 10^{-3})(7.152 \times 10^{3})$$
$$= -33.38\ V/V$$

$$\rightarrow \alpha = \frac{R_o}{R_s + R_o} = \frac{68.99k}{2k + 68.99k} = 0.9718$$

$$A_v = \frac{V_{out}}{V_{in}} = \alpha A_{vo} = 0.9718 \times (-33.32)$$
$$= -32.44\ V/V$$

$$|A_v|_{dB} = 30.22\ dB \geq 26.02\ dB$$

$$f_L = \frac{1}{2\pi (R_s + R_o) C_c} = \frac{1}{2\pi (2k + 68.99k) \times 2.50\ pF}$$

$$= 896.8\ kHz = 0.90\ MHz \leq 1\ MHz$$

$C_{ov,n} = C_{ov} \times W_n = 0.35 \times 20.8\,\mu m = 7.28\,fF$

$C_{ox,n} = W_n L_n C_{ox} = 20.8 \times 0.3 \times 18\,fF = 112.32\,fF$

$C_{gs,n} = \frac{2}{3} C_{ox,n} = \frac{2}{3}(112.32) = 82.16\,fF$

$C_{gd,n} = C_{ov,n} = 7.28\,fF$

$C_{db,n} = C_{db} \times W_n = 0.5 \times 20.8 = 10.90\,fF$

$C_{gd,p} = C_{ov} \times W_p = 0.35 \times 22.5 = 7.88\,fF$

$C_{db,p} = C_{db} \times W_p = 0.5 \times 22.5 = 11.25\,fF$

$C_{out,tot} = C_{gd,n} + C_{db,n} + C_{gd,p} + C_{db,p} + C_L + C_2$

$C_{out,tot} = 7.28 + 10.90 + 7.88 + 11.25 + 9.81 + 100$

$= 141.62\,fF$

$f_{p,out} = \frac{1}{2\pi R_{out} \cdot C_{out,tot}} = \frac{1}{2\pi \times 7.152k \times 141.62}$

$= 157.1\,MHz$

$C_{in,miller} = C_{gs,n} + C_{gd,n}(1 + |A_v|) = 82.16 + 7.28(1 + 22\pi)$

$= 256.7\,fF$

$f_{p,in} = \frac{1}{2\pi \cdot R_{in,eq} \cdot C_{in,miller}} = \frac{1}{2\pi \times 1.944k \times 256.7f}$

$= 319.0\,MHz$

$f_H \approx \frac{1}{\sqrt{\frac{1}{f_{p,out}^2} + \frac{1}{f_{p,in}^2}}} = 141.0\,MHz \geq 100\,MHz$

Design -3.

① Dc operating Point & Sizing Summary.

$I_{D,amp} = 181\,\mu A$, $I_{REF} = 50\,\mu A$

$W_N/L_N = 20.8/0.3$ (M_0)

$W_P/L_P = 22.5/0.9$ (M_1)

$W_{N_1}/L_{N_1} = W_{N_2}/L_{N_2} = 5.78/0.3$

$W_{P_1}/L_{P_1} = 6.25/0.4$

② Gain analysis

$$g_{m,n} = \frac{2I_D}{V_{ov,n}} = \frac{2\times 181}{0.18} = 2.01\,mS$$

$$r_{o,n} = \frac{1}{\partial_N I_D} = \frac{1}{0.3367\times 181} = 16.4\,k\Omega$$

$$r_{o,p} = \frac{1}{\partial_P I_D} = \frac{1}{0.1600\times 181} = 34.5\,k\Omega$$

$$R_{out} = r_{o,n} \parallel r_{o,p} = 11.1\,k\Omega$$

$$|A_v| = g_{m,n} R_{out} = 2.01\,mS \times 11.1\,k\Omega$$

$$= 26.97\,dB$$

(8) Cutoff frequencies:

$$f_H = \frac{1}{2\pi R_{out} C_{L,tot}} = \frac{1}{2\pi \times 11.1 \times 120.} = 119.5MH$$

$$f_H \geq 100\,MHz$$

$$f_L = \frac{1}{2\pi (R_S + R_{bias}) C_{in}} = \frac{1}{2\pi (2 + 68.98) \times 2.5}$$

$$= 877\,kHz$$

$$f_L \leq 877kHz$$

Virtuoso (R) Visualization & Analysis XL: assignment2 CS_amplifier maestro

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cadence

assignment2_CS_amplifier_1

Gain_dB

Thu Oct 15 21:53:06 20...

Name | Vis

Gain_dB

M7: 10.0MHz 26.90972dB

M9: 1.0MHz 23.90972dB

M8: 100.0MHz 23.90972dB

25.0

0.0

-25.0

-50.0

-75.0

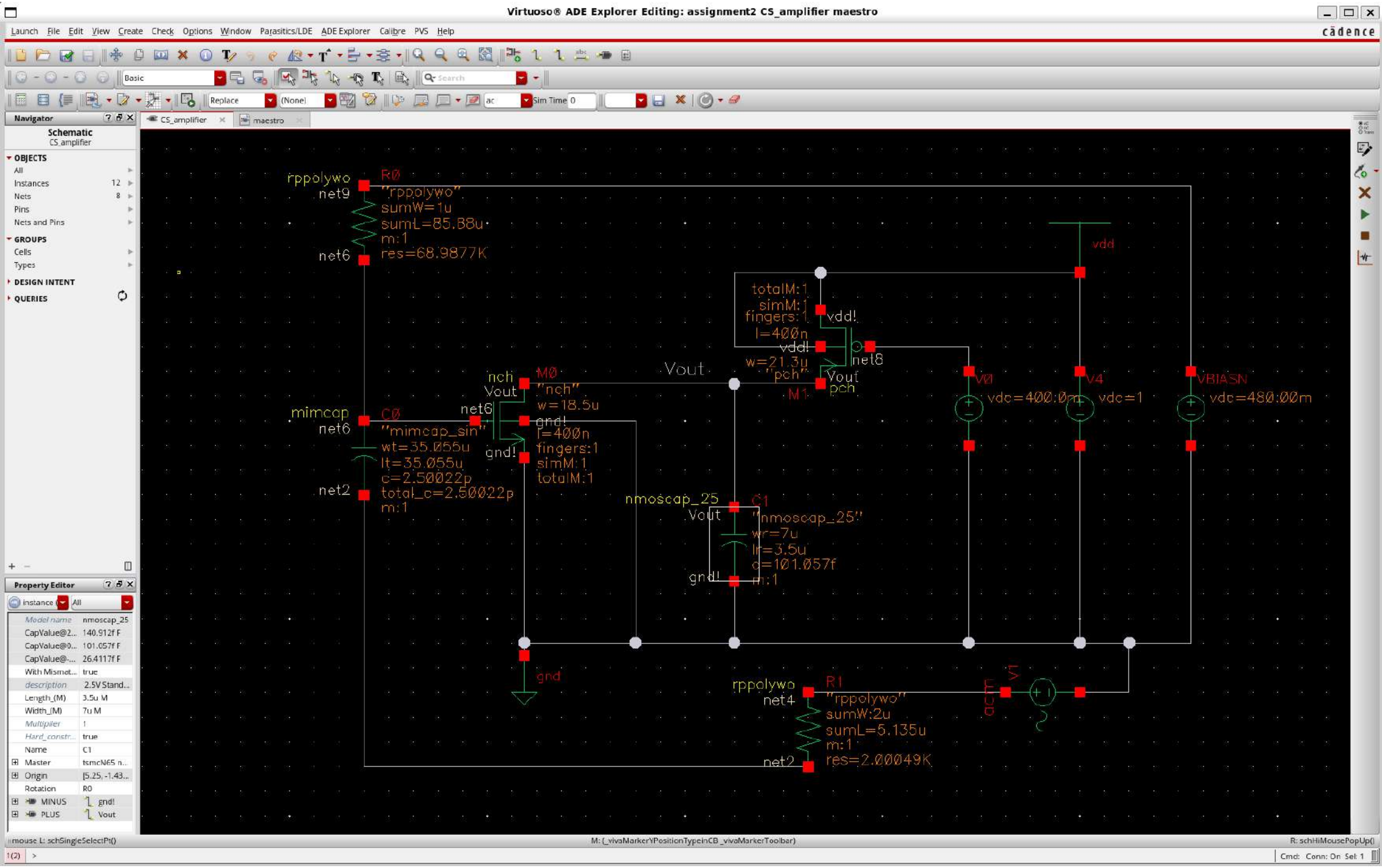
Y (dB)

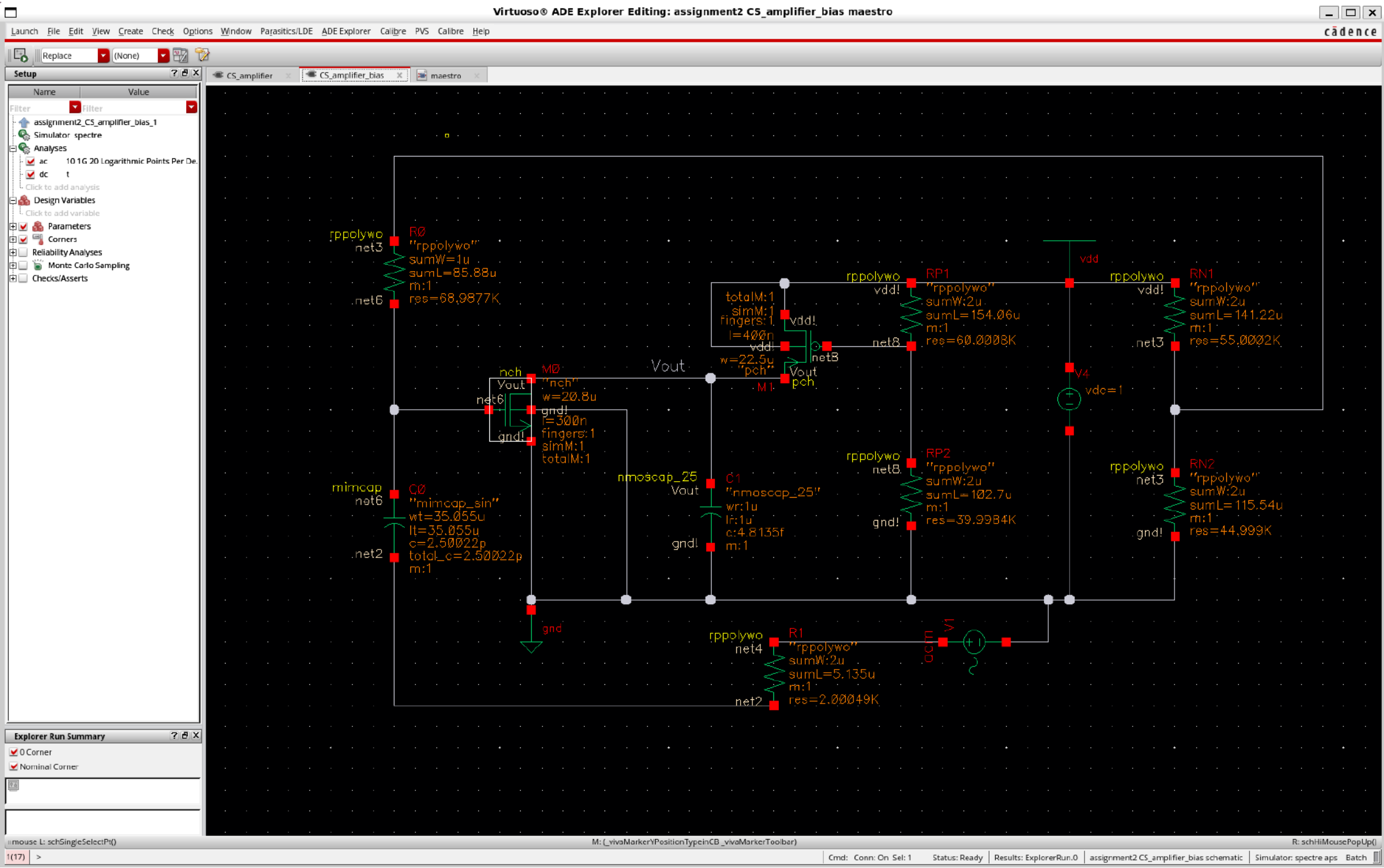
10^1 10^2 10^3 10^4 10^5 10^6 10^7 10^8 10^9

freq (Hz)

mouse L: M: R:

5(14) Trace: Gain_dB; History: ExplorerRun.0; Test: assignment2_CS_amplifier_1


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CS_amplifier
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All
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Nets 8
Pins
Nets and Pins
GROUPS
Cells
Types
DESIGN INTENT
QUERIES
Property Editor
CS_amplifier
maestro
rppolywo
net9
R0
"rppolywo"
sumW=1u
sumL=85.88u
m:1
res=68.9877K
net6
totalM:1
simM:1
fingers:1
l=400n
vdd!
w=21.3u
"pch"
M1
net8
Vout
vdd
V0
vdc=400.0m
V4
vdc=1
VBIASN
vdc=480.00m
nch
M0
"nch"
w=18.5u
gnd!
l=400n
fingers:1
simM:1
totalM:1
mimcap
net6
C0
"mimcap_sin"
wt=35.055u
lt=35.055u
c=2.50022p
total_c=2.50022p
m:1
net2
nmoscap_25
C1
"nmoscap_25"
wr=7u
lr=3.5u
c=101.057f
m:1
gnd
rppolywo
net4
R1
"rppolywo"
sumW:2u
sumL=5.135u
m:1
res=2.00049K
V1
dc=1
Model name nmoscap_25
CapValue@2... 140.912f F
CapValue@0... 101.057f F
CapValue@-... 26.4117f F
With Mismat... true
description 2.5V Stand...
Length_(M) 3.5u M
Width_(M) 7u M
Multiplier 1
Hard_constr... true
Name C1
Master tsmcN65 n...
Origin [5.25, -1.43...
Rotation R0
MINUS gnd!
PLUS Vout
mouse L: schSingleSelectPt()
M: (_vivaMarkerYPositionTypeinCB _vivaMarkerToolbar)
R: schHiMousePopUp()
Cmd: Conn: On Sel: 1


Virtuoso® ADE Explorer Editing: assignment2 CS_amplifier_bias maestro
Launch File Edit View Create Check Options Window Parasitics/LDE ADE Explorer Calibre PVS Calibre Help
cadence
Replace (None)
Setup
Name Value
assignment2_CS_amplifier_bias_1
Simulator spectre
Analyses
ac 10 1G 20 Logarithmic Points Per De.
dc t
Design Variables
Parameters
Corners
Reliability Analyses
Monte Carlo Sampling
Checks/Asserts
Explorer Run Summary
0 Corner
Nominal Corner
CS_amplifier
CS_amplifier_bias
maestro
rppolywo
net3
R0
"rppolywo"
sumW=1u
sumL=85.88u
m:1
res=68.9877K
net6
vdd
rppolywo
vdd!
RP1
"rppolywo"
sumW:2u
sumL=154.06u
m:1
res=60.0008K
net8
totalM:1
simM:1
fingers:1
l=400n
vdd!
w=22.5u
"pch"
M1
pch
net8
Vout
rppolywo
vdd!
RN1
"rppolywo"
sumW:2u
sumL=141.22u
m:1
res=55.0002K
net3
V4
vdc=1
nch
Vout
net6
gnd!
M0
"nch"
w=20.8u
l=300n
fingers:1
simM:1
totalM:1
Vout
nmoscap_25
Vout
gnd!
C1
"nmoscap_25"
wr:1u
lr:1u
c:4.8135f
m:1
rppolywo
net8
RP2
"rppolywo"
sumW:2u
sumL=102.7u
m:1
res=39.9984K
gnd!
rppolywo
net3
RN2
"rppolywo"
sumW:2u
sumL=115.54u
m:1
res=44.999K
gnd!
mimcap
net6
C0
"mimcap_sin"
wt=35.055u
lt=35.055u
c=2.50022p
total_c=2.50022p
m:1
net2
gnd
rppolywo
net4
R1
"rppolywo"
sumW:2u
sumL=5.135u
m:1
res=2.00049K
net2
dc=0
V1
mouse L: schSingleSelectPt()
M: (_vivaMarkerYPositionTypeinCB _vivaMarkerToolbar)
R: schHiMousePopUp()
1(17)
Cmd: Conn: On Sel: 1
Status: Ready
Results: ExplorerRun.0
assignment2 CS_amplifier_bias schematic
Simulator: spectre aps Batch

Virtuoso (R) Visualization & Analysis XL: assignment2 CS_amplifier_bias maestro

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cadence

assignment2_CS_amplifier_bias_1

Gain_dB

Thu Oct 15 23:32:48 20... 1

Name | Vis

Gain_dB

M16: 1.0MHz 25.87558dB

M17: 5.623413MHz 27.22374dB

M15: 100.0MHz 23.70656dB

25.0

0.0

-25.0

-50.0

-75.0

Y (dB)

10^1 10^2 10^3 10^4 10^5 10^6 10^7 10^8 10^9

freq (Hz)

mouse L: M: R:

33(101) Save Window Ctrl+S

Virtuoso (R) Visualization & Analysis XL: assignment2 CS_amplifier_current_mirror maestro

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Subwindow: 1 Position: x y Data Point Classic

assignment2_CS_amplifier_current_...

Gain_dB

Fri Oct 16 20:19:32 2026

Name | Vis

Gain_dB

M3: 1.0MHz 25.93178

M1: 12.58925MHz 28.17065

M2: 100.0MHz 27.12719

25.0

0.0

-25.0

-50.0

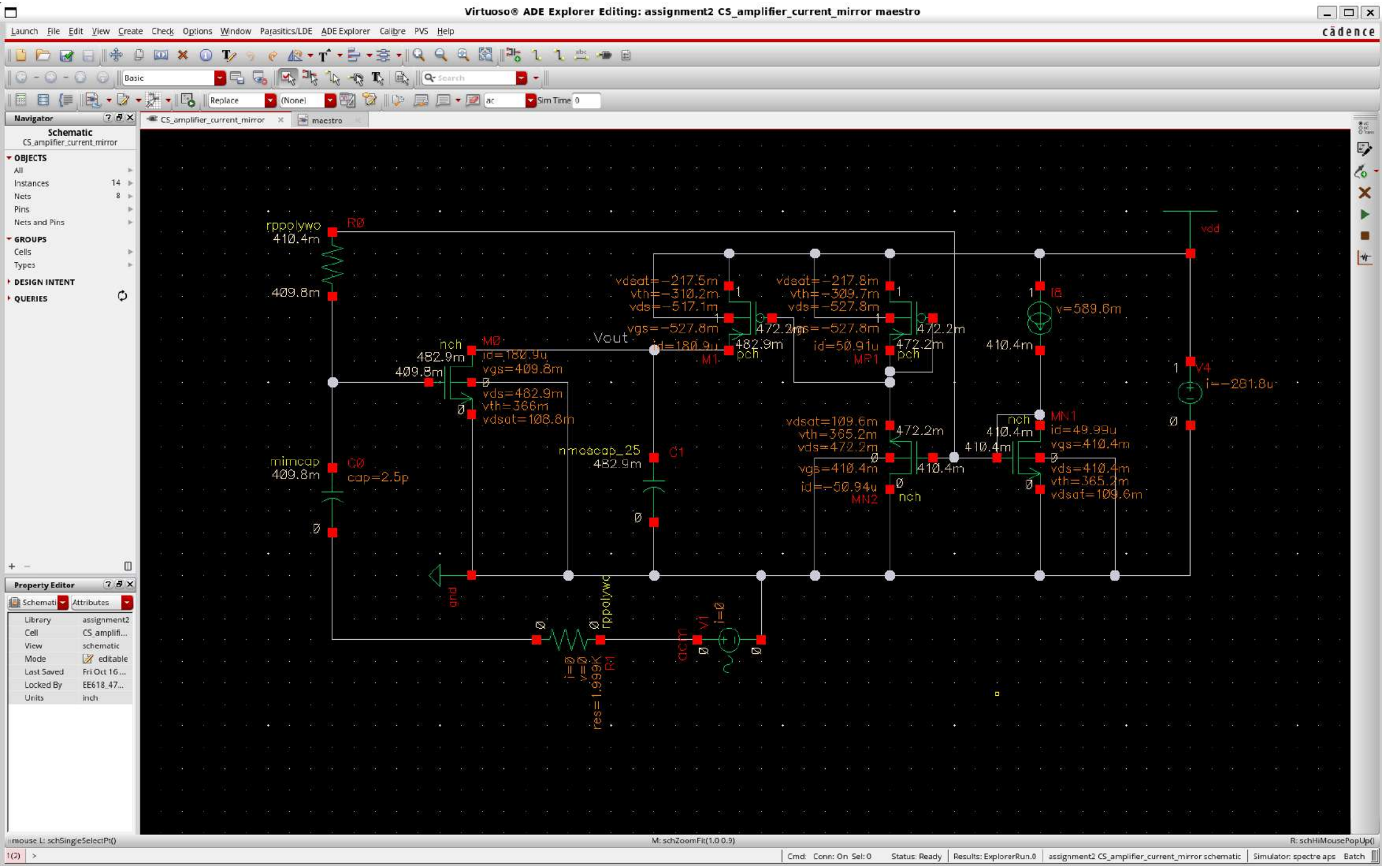
-75.0

10^1 10^2 10^3 10^4 10^5 10^6 10^7 10^8 10^9

freq (Hz)

mouse L: M: R:

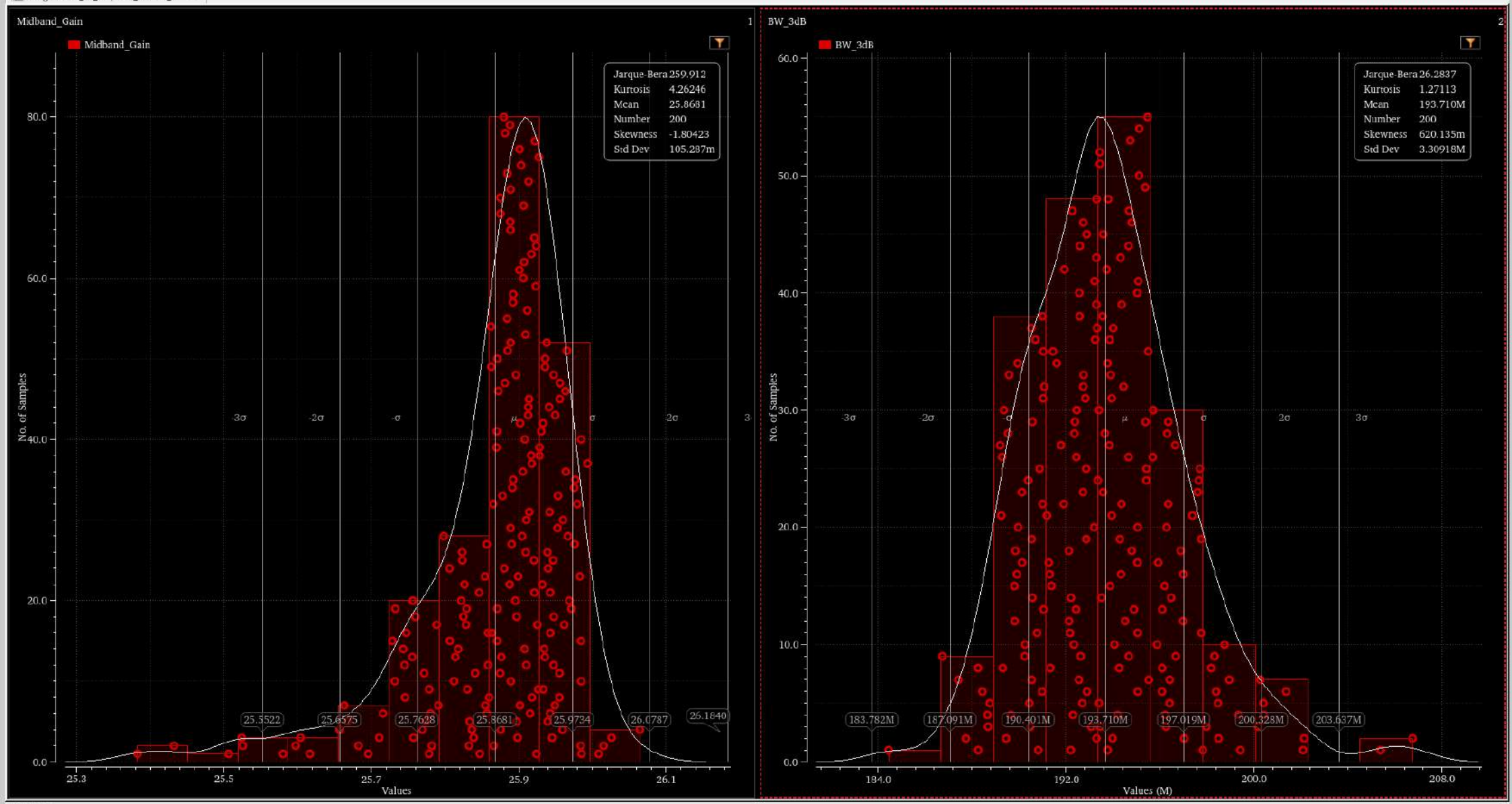
4(15) no error


Virtuoso® ADE Explorer Editing: assignment2 CS_amplifier_current_mirror maestro
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GROUPS
Cells
Types
DESIGN INTENT
QUERIES
CS_amplifier_current_mirror
maestro
rppolywo
410.4m
R0
409.8m
nch
M0
482.9m
id=180.9u
409.8m
vgs=409.8m
vds=482.9m
vth=366m
vdsat=106.8m
Vout
vdsat=-217.5m
vth=-310.2m
vds=-517.1m
vgs=-527.8m
id=180.9u
M1
482.9m
pch
vdsat=-217.8m
vth=-309.7m
vds=-527.8m
vgs=-527.8m
id=50.91u
MP1
472.2m
pch
472.2m
I8
v=589.6m
410.4m
vdd
V4
i=-281.8u
mimcap
409.8m
C0
cap=2.5p
nmoscap_25
482.9m
C1
vdsat=109.6m
vth=365.2m
vds=472.2m
vgs=410.4m
id=-50.94u
MN2
472.2m
nch
410.4m
nch
410.4m
MN1
id=49.99u
vgs=410.4m
vds=410.4m
vth=365.2m
vdsat=109.6m
gnd
rppolywo
R1
res=1.999K
i=0
v=0
acm
V1
i=0
Property Editor
Schemati
Attributes
Library assignment2
Cell CS_amplifi...
View schematic
Mode editable
Last Saved Fri Oct 16 ...
Locked By EE618_47...
Units inch
mouse L: schSingleSelectPt()
M: schZoomFit(1.0 0.9)
R: schHiMousePopUp()
Cmd: Conn: On Sel: 0
Status: Ready
Results: ExplorerRun.0
assignment2 CS_amplifier_current_mirror schematic
Simulator: spectre aps Batch

Virtuoso (R) Visualization & Analysis XL: assignment2 CS_amplifier_current_mirror maestro

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cadence



plot new graph subwindow

Midban...	100% (200/2...	25.38	info	26.06	25.87	105.3m	0
BW_3dB	100% (200/2...	184.5M	info	206.7M	193.7M	3.309M	0